

MAT1002 Lecture 23,

Outline

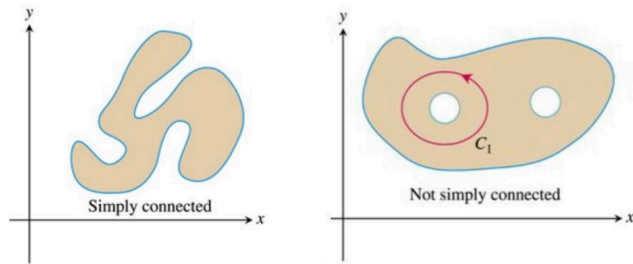
- Component tests
- Green's theorem

Component Test

“no hole”

A connected plane region D is said to be simply connected if every simple closed curve in D encloses only points that are in D .

Another way to think about it :
a simple closed curve in D can be “shrunk” to a point without leaving D .
This can be generalised to 3D.



Theorem (Component Test for Conservative Fields) (in $\mathbb{R}^2$)

Let $\mathbf{F}(x, y) := \langle M(x, y), N(x, y) \rangle$ be a vector field on an open simply connected domain D , such that M and N have continuous partial derivatives. Then $\mathbf{F}$ is conservative on D if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

on D .

Example

The vector field defined by $\mathbf{F}(x, y) := \langle -y, x \rangle$ is not conservative on $\mathbb{R}^2$, since it does not pass the component test.

Problem

Consider the vector field defined by

$$\mathbf{F}(x, y) := \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle.$$

We have shown that $\mathbf{F}$ is not conservative. On the other hand, one can check that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

This seems to contradict the component test. What is wrong here?

Example

Consider the vector field defined by

$$\mathbf{F}(x, y) := \langle 3 + 2xy, x^2 - 3y^2 \rangle.$$

- (a) Show that $\mathbf{F}$ is conservative on $\mathbb{R}^2$.
- (b) Find a potential function for $\mathbf{F}$.
- (c) Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is parametrized by

$$\mathbf{r}(t) = \langle e^t \sin t, e^t \cos t \rangle, \quad 0 \leq t \leq \pi.$$

Ans: (b) $f(x, y) = 3x + x^2y - y^3 + C$
(c) $e^{3\pi} + 1$.

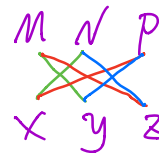
Component Test in 3D

There is also a component test for vector fields $\mathbf{F} := \mathbf{F}(x, y, z)$ in $\mathbb{R}^3$. Under assumptions similar to those on Page 25, $\mathbf{F}$ is conservative if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

for the 2D component test

M, N, P are on a simply connected region



Exact Differential Form

Consider $\vec{F} = \langle M, N, P \rangle$ (3D vector field). Then

- $\int_C \vec{F} \cdot d\vec{r} = \int_C Mdx + Ndy + Pdz$.
- If $\vec{F} = \nabla f$ is a gradient field, then

$$\int_C Mdx + Ndy + Pdz = \int_C \underbrace{f_x dx + f_y dy + f_z dz}_{\text{total differential } df}$$

Def: • An expression $Mdx + Ndy + Pdz$ is a **differential form**.
• A differential form is **exact** on D if it is the total differential df for some real-valued (i.e. scalar) function f (on D).

Remark: $Mdx + Ndy + Pdz$ is exact if and only if $\vec{F}$ is conservative (with potential function f).

We summarize some of the key results of this week as a theorem.

Theorem

Let $\mathbf{F}$ be a vector field whose components are continuous on an open connected region D . Then the following conditions are equivalent.

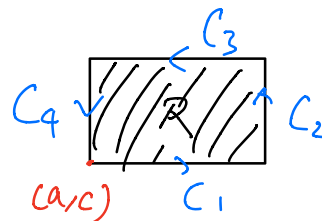
- (a) The field $\mathbf{F}$ is conservative on D .
- (b) There exists a function f such that $\mathbf{F} = \nabla f$ on D .
- (c) For every closed curve C in D , $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

Simply Connected D

Under certain assumptions, whether (a) holds can be tested using the component test.

Green's Theorem

$$= \oint_C \vec{F} \cdot \vec{T} ds$$



- Given the velocity field $\vec{F}$ of some fluid, consider the counterclockwise circulation along the boundary C of rectangle $R := [a,b] \times [c,d]$

$$\oint_C \vec{F} \cdot \vec{T} ds = \oint_C Mdx + \oint_C Ndy$$

$$\begin{aligned} \oint_C Mdx &= \int_{C_1} Mdx + \int_{C_2} Mdx + \int_{C_3} Mdx + \int_{C_4} Mdx \\ &= \int_{C_1} Mdx - \int_{-C_3} Mdx \end{aligned}$$

$\overset{=0}{\text{since } dx=0}$

$$(C_1: \vec{r}_1(t) = \langle t, c \rangle, a \leq t \leq b; -C_3: \vec{r}_3(t) = \langle t, d \rangle, a \leq t \leq b.)$$

$$= \int_a^b M(t, c) dt - \int_a^b M(t, d) dt$$

$$\begin{aligned}
&= \int_a^b [M(t,c) - M(t,d)] dt \stackrel{\text{FTC}}{=} \int_a^b \int_c^d \frac{\partial M}{\partial y}(t,y) dy dt \\
&= \int_a^b \int_c^d \frac{\partial M}{\partial y}(x,y) dy dx \quad (\text{Rename } t \text{ to } x) \\
&= - \int_a^b \int_c^d \frac{\partial M}{\partial y}(x,y) dy dx \\
&= - \iint_R \frac{\partial M}{\partial y}(x,y) dA \quad \textcircled{1}
\end{aligned}$$

• Similarly,

$$\begin{aligned}
\oint_C N dy &= \int_{C_2} N dy - \int_{-C_4} N dy & C_2: \vec{r}_2(t) = \langle b, t \rangle, \quad c \leq t \leq d \\
& & -C_4: \vec{r}_4(t) = \langle a, t \rangle, \quad c \leq t \leq d \\
&= \int_c^d N(b,t) dt - \int_c^d N(a,t) dt
\end{aligned}$$

$$\begin{aligned}
&\stackrel{\text{FTC}}{=} \int_c^d \int_a^b \frac{\partial N}{\partial x}(x,t) dx dt = \int_c^d \int_a^b \frac{\partial N}{\partial x}(x,y) dx dy \\
&= \iint_R \frac{\partial N}{\partial x}(x,y) dA. \quad \textcircled{2}
\end{aligned}$$

• Combining ① & ②, we have

$$\oint_C \vec{F} \cdot \vec{T} ds = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA. \quad \textcircled{3}$$

• You can use the same technique to show that ③ also holds if R is a region that is both of type-I and type-II. In fact, the result holds for any bounded region R that has a simple closed boundary, as stated formally below.

Theorem (Green's Theorem) (Circulation Version)

Let C be a ~~positively oriented~~ ^{counterclockwise} piecewise-smooth, simple closed curve in the xy -plane, and let R be the region bounded by C . Assume that M and N have continuous partial derivatives on an open region containing R . Then

$$\oint_C \vec{F} \cdot \vec{T} ds = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

C is the boundary of R .

A result about counterclockwise circulation.

Key Green's theorem (above) relates a double integral $\iint_R$ with a circulation integral along its counterclockwise BOUNDARY C .

e.g. Find the clockwise circulation of $\vec{F} := \langle x^2, xy \rangle$ along the triangle with vertices $(0,0)$, $(1,0)$ and $(0,1)$.

Ans: $-\frac{1}{6}$.

Example

Evaluate

$$\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy,$$

Ans: 36π

where C is the circle $x^2 + y^2 = 9$, traversed ~~clockwise~~ counterclockwise.

Exercise

Evaluate

$$\oint_C y^2 dx + 3xy dy,$$

where C is the positively oriented boundary of the region R given by $1 \leq x^2 + y^2 \leq 4$, $y \geq 0$.

Ans: $14/3$.

Recall the component test.

Theorem (Component Test for Conservative Fields) (in $\mathbb{R}^2$)

Let $\mathbf{F}(x, y) := \langle M(x, y), N(x, y) \rangle$ be a vector field on an open simply connected domain D , such that M and N have continuous partial derivatives. Then $\mathbf{F}$ is conservative on D if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

on D .

We can now prove it.

Proof: " $\Rightarrow$ ": Assume that $\vec{F}$ is conservative on D . Then

$\exists f$ s.t. $\vec{F} = \nabla f = \langle f_x, f_y \rangle$ on D . Now

$$\frac{\partial M}{\partial y} = (f_x)_y = f_{xy} = f_{yx} = (f_y)_x = \frac{\partial N}{\partial x}.$$

$\uparrow$ Mixed derivative theorem.

" $\Leftarrow$ ": Assume $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ on D .

- Let C be any closed curve in D . First assume C is simple.
- Since D is simply connected, the region R bounded by C lies entirely in D .
- By Green's theorem, $\oint_C \vec{F} \cdot \vec{T} ds = \iint_R \underbrace{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)}_{\equiv 0 \text{ on } R \subseteq D} dA = 0$
- If C is a closed curve which is not simple, decompose it into simple closed curves.

- Hence, $\oint_C \vec{F} \cdot \vec{T} ds = 0$, $\forall$ closed curve C in D .
- By the loop property, $\vec{F}$ is conservative on D . □

We have a similar result for flux (in place of circulations).

- Consider the flux across the boundary of a rectangle R .
- From last time, $\text{flux} = \oint_C \vec{F} \cdot \vec{n} ds = \oint_C M dy - N dx$.
- Using the reasoning on Page 5, we have

$$\oint_C M dy = \iint_R \frac{\partial M}{\partial x} dA \quad \& \quad \oint_C N dx = - \iint_R \frac{\partial N}{\partial y} dA$$

- Hence, $\text{flux} = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA$.

The ^{i.e., R may not be rectangle} general result is given by the following theorem.

Theorem (Green's Theorem, Flux Version)

Under the assumptions for Green's theorem (circulation version), except orientation (direction) of C is no longer required,

$$\oint_C \vec{F} \cdot \vec{n} ds = \oint_C M dy - N dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA.$$

Question

- Consider $\mathbf{F}(x, y) := \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$.
- Let C be the unit circle $x^2 + y^2 = 1$.
- We shown that $\oint_C \mathbf{F} \cdot \vec{T} ds = 2\pi$ by direct computation.
- On the other hand, one can check that $\frac{\partial m}{\partial y} = \frac{\partial n}{\partial x}$ always holds.
- By Green's theorem, $\oint_C \mathbf{F} \cdot \vec{T} ds = \iint_R \left(\frac{\partial n}{\partial x} - \frac{\partial m}{\partial y} \right) dA = 0$.
- Therefore, $2\pi = 0$. 

O.K., seriously, what is wrong?